

Team RUN

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COMP 491, API Design & System Workflow assignment

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API Design & System Workflow assignment

Part 1: API Design

1. Register User:

API Name:

- Register User

Method:

- POST

Input:

- username, email, password, heroName

Output:

- success message, user ID, basic user profile

Module:

- User/Authentication Module

Related Database Table(s):

- Users

Explanation:

This API is used when a new user creates an account. The system takes the users information, saves it in the database and creates their profile.

2. Login User

API Name:

- Login User

Method:

- POST

Input:

- email, password

Output:

- login success message, user profile information, hero name, gold

Module:

- User/Authentication Module

Related Database Collections:

- Users

Explanation

This API checks if the user's email and password are correct. If the login works, the system returns a token so the user can access protected parts of the app.

3. Get Player Profile

API Name:

- Get Player Profile

Method:

- GET

Input:

- user ID

Output:

- hero name, level, XP, strength, stamina, agility

Module:

- Profile Module

Related Database Collections:

- Users, Characters

Explanation

This API loads the player's profile information. It is used when the system needs to show the user's level, stats, and overall progress.

4. Log Workout

API Name:

- Log Workout

Method:

- POST

Input:

- exercise name, body part, reps, sets, weight

Output:

- saved workout, XP gained, updated character stats

Module:

- Workout Module

Related Database Collections:

- Workouts, Characters, Users

Explanation

This API saves workout information in the database. It also updates the user's character because the app connects real workouts to RPG progress.

5. Log Water Intake

API Name:

- Log Water Intake

Method:

- POST

Input:

- amount of water or other intake

Output:

- updated hydration progress, reward update

Module:

- Hydration Module

Related Database Collections:

- WaterLogs, Users

Explanation

This API records how much water the user drinks. It helps the system keep track of hydration goals and connect them to the character's progress.

6. Get Dashboard Summary

API Name:

- Get Dashboard Summary

Method:

- GET

Input:

- authenticated user ID

Output:

- workout summary, hydration summary, recent activity, level progress

Module:

- Dashboard Module

Related Database Collections:

- Workouts, WaterLogs, Characters, Users

Explanation

This API collects the main information the user should see on the dashboard. It brings together data from different parts of the system and displays a summary.

7. Start Battle

API Name:

- Start Battle

Method:

- POST

Input:

- authenticated user ID

Output:

- battle start information, enemy data, user stats

Module:

- Battle Module

Related Database Collections:

- Battles, Characters, Users

Explanation

This API begins a battle in the RPG part of the app. It uses the player's current stats to prepare the battle.

8. End Battle

API Name:

- End Battle

Method:

- POST

Input:

- battle ID, result, remaining health

Output:

- XP reward, gold reward, battle result

Module:

- Battle Module

Related Database Collections:

- Battles, Characters, Users

Explanation

This API saves the result of the battle. It also updates the rewards the user earns, such as XP or gold.

9. Get Leaderboard

API Name:

- Get Leaderboard

Method:

- GET

Input:

- none

Output:

- top players ranked by XP, level, or wins

Module:

- Leaderboard Module

Related Database Collections:

- Users, Characters

Explanation

This API gets ranking information so users can compare their progress with others. It adds more motivation and competition to the app.

10. AI Workout Recommendation

API Name:

- AI Workout Recommendation

Method:

- POST

Input:

- workout history, body parts trained, hydration history, current stats

Output:

- suggested workout or recommended next exercise

Module:

- AI Recommendation Module

Related Database Collections:

- Workouts, WaterLogs, Characters, Users

Explanation

This API is used for the AI feature. It looks at the users past activity and gives a workout suggestion to help them know what to do next.

Part 2: API Grouping

User / Authentication Module

Register User

Login User

Get Player Profile

Explanation

These APIs are grouped together because they all deal with user identity, login, and account information.

Workout Module

Log Workout

Explanation

This API belongs here because it handles workout input and character progress updates.

Hydration Module

Log Water Intake

Explanation

This API is grouped here because it only handles hydration tracking.

Dashboard Module

Get Dashboard Summary

Explanation

This API belongs to the dashboard because it loads the summary information shown on the main screen.

Battle Module

Start Battle

End Battle

Explanation

These APIs are grouped together because they both belong to the battle system.

Leaderboard Module

Get Leaderboard

Explanation

This API is separated because it handles ranking and comparison between users.

AI Recommendation Module

AI Workout Recommendation

Explanation

This API belongs to the AI module because it generates recommendations instead of only saving or retrieving normal data.

Part 3: System Workflow

Feature: Log Workout

1. The user opens the Log Workout screen.
2. The user enters the workout information such as exercise name, reps, sets, weight, and body part.
3. The UI sends a POST request to the Log Workout API.
4. The backend receives the request and checks if the input is valid.
5. The backend calculates XP based on the workout details.
6. The workout data is saved into the Workouts collection.
7. The users character stats are updated in the database.
8. The backend sends the updated XP and stats back to the UI.
9. The dashboard then shows the new progress.

Explanation

This workflow shows how one action in the app moves through the whole system. The user starts in the UI, the request goes through the API, the backend processes it, the database is updated, and then the result is shown back to the user.

Part 4: AI Integration

AI Feature: Workout Recommendation System

Which API is used

- AI Workout Recommendation API

Input

- workout history
- body parts trained
- hydration history
- current character stats

Output

- suggested workout
- recommended exercise

Where it appears in UI

- Dashboard screen

Explanation

The AI feature looks at what the user has already done and tries to suggest a helpful next workout. This makes the app feel more useful because the user does not have to guess what to do next.

Part 5: Error Handling

Invalid Input

Example: user enters negative reps or invalid weight amount

System Response: show error message like "Invalid workout input"

Explanation

This helps stop incorrect information from being saved in the system.

Missing Data

Example: user submits workout without name, reps, or body part

System Response: show error message "Missing required fields"

Explanation

This makes sure the API only saves complete data.

System Error

Example: database connection fails or server crashes

System Response: show error message "Server error, please try again"

Explanation

This lets the user know the issue is on the system side and not because they entered something wrong.